Yaoxuan (Thaddy) Wu

Education

University of California, Los Angeles — **Los Angeles, CA**
Ph.D. Student — *Sep. 2022 – Present*

University of California, Los Angeles — **Los Angeles, CA**
M.S. — *Sep. 2022 – Mar. 2025*

Peking University — **Beijing, China**
B.S., Turing Class, Summa Cum Laude — *Sep. 2018 – Jun. 2022*

Internship

Amazon Web Services — **Arlington, VA**
Research Scientist Intern — *Jun. 2025 – Sep. 2025*

Research Overview

- **AI Systems.** I develop AI techniques for software engineering by combining large language models with symbolic reasoning and property-based methods. My work spans LLM-based test generation, automated proof synthesis, and cloud system simulation, and is expanding toward AI-native software systems and autonomous software agents. Representative projects include *PALM* (*ICPC 2026*), *DualVeri (ICSE 2027)* , and *CloudTwin* (Preprint, 2026).
- **Testing and Verification for DISC Systems.** I develop testing and verification techniques for Apache Spark and other data-intensive scalable computing (DISC) systems using symbolic execution, fuzzing, and property-based testing. Representative projects include *NaturalSym* (*FSE 2024*), *NaturalFuzz* (*ASE 2023*), *Holey* (Preprint, 2026), and the position paper *Why Property-Based Testing* (*FSE DISE Workshop 2026*).

Publications

* Equal contribution.

ICSE'27 DualVeri: Agentic Proof and Property-Based Testing via Property-Templates in Data-Intensive Computing.
Seongmin Lee*, **Yaoxuan Wu***, and Miryung Kim. arXiv:2607.09072 (2026).

Preprint Properties with Holes: Meta-Property-Based Testing of Data-Intensive Computing Frameworks.
Yaoxuan Wu, Ingrid Lee, Ahmad Humayun, Muhammad Ali Gulzar, and Miryung Kim. arXiv:2606.11132 (2026).

Preprint CloudTwin: Inverted Tool Use for Long-Horizon Cloud Simulation.
Yifan Zhang*, **Yaoxuan Wu***, Leyi Cui*, Hui Guan, Brandon Paulsen, Joey Dodds, and Daniel Kroening.

Preprint DAG-based Workload Generation for Testing Data-Intensive Scalable Computing Frameworks.
Ahmad Humayun, **Yaoxuan Wu**, Muhammad Ali Gulzar, and Miryung Kim.

FSE'26 WS Why Property-Based Testing is Necessary for Data-Intensive Scalable Computing.
Yaoxuan Wu, Ingrid Lee, Ahmad Humayun, Muhammad Ali Gulzar, and Miryung Kim.

ICPC'26 PALM: Path-aware LLM-based Test Generation with Comprehension.
Yaoxuan Wu, Xiaojie Zhou, Ahmad Humayun, Muhammad Ali Gulzar, and Miryung Kim.

FSE'24 Natural Symbolic Execution-based Testing for Big Data Analytics.
Yaoxuan Wu, Ahmad Humayun, Muhammad Ali Gulzar, and Miryung Kim.

ASE'23 NaturalFuzz: Natural Input Generation for Big Data Analytics.
Ahmad Humayun, **Yaoxuan Wu**, Miryung Kim, and Muhammad Ali Gulzar.

USENIX'23 Pelican: Exploiting Backdoors of Naturally Trained Deep Learning Models in Binary Code Analysis.
Zhuo Zhang, Guanhong Tao, Guangyu Shen, Shengwei An, Qiuling Xu, Yingqi Liu, Yapeng Ye, **Yaoxuan Wu**, and Xiangyu Zhang.

Preprint The Parser Already Knows: Lightweight Bias Correction in Constrained Decoding.
Isil Özgü, **Yaoxuan Wu**, Guy Van den Broeck, and Miryung Kim. arXiv:2608.10137 (2026).

Preprint Targeted Testing of Compiler Optimizations via Grammar-Level Composition Styles.
Zitong Zhou, Ben Limpanukorn, Hong Jin Kang, Jiyuan Wang, **Yaoxuan Wu**, Ákos Kiss, Renata Hodovan, and Miryung Kim. arXiv:2512.04344 (2025).

Awards

I am an Amazon Fellowship recipient and have received multiple scholarships. During high school and university, I competed in algorithmic programming contests, earning a gold medal at the National Olympiad in Informatics (NOI), China, and multiple gold and silver medals at ICPC Asia regional contests and the East Continent Final.